

Marikina LegisHub: An E-Governance-Based Legislative Document Management System for Enhanced Efficiency, Transparency, and Data Integrity

Software Requirements Specification

- Version 1 -

03 | 14 | 2026

Prepared for
ASST. PROF. CAMILO P. ABOGADO

Revision History

Date	Description	Author	Comments
03 / 31 / 26	Version 1	Jake Hasley Hafalla and Mylene Esquilona	Initial Draft
04 / 05 / 26	Version 2	Mylene G. Esquilona	Added description of the modules
04 / 06 / 26	Version 3	Mylene G. Esquilona	Added section 3.4 Class / Objects, 3.5.1.1 Attributes and 3.5.1.2 Functions
04 / 10 / 26	Version 4	Mylene G. Esquilona	Added UI and Hardware Interface in 2.1 External Interface Requirements. Changed the format of section 3.1, 3.2 and 3.3.
04 / 11 / 26	Version 4.1	Mylene G. Esquilona	Integrated Section 3.3 Functional Validations (F1-F49) and merged old validation rules.
04 / 16 / 26	Version 5	Mylene G. Esquilona	Finalized Section 3.4 Classes/Objects and 3.5.1.1 Attributes
04 / 17 / 26	Version 6	Mylene G. Esquilona	Updated Section 3.9 Other

			Requirements (API/Libraries)
--	--	--	---------------------------------

Document Approval

The following Software Requirements Specification has been accepted and approved by the following:

Printed Name	Title	Date
Jake Hasley Hafalla	Project Manager	03/31/26
Jayvee Moral Gerald Sy	Front/Backend Developer	04/05/26
Jake Hasley Hafalla	Project Manager	04/08/26
Myra Luna Gaurino	UI/UX	04/11/26
Jake Hasley Hafalla	Project Manager	04/12/26
Jayvee Moral Gerald Sy	Front/Backend Developer	04/16/26
Jake Hasley Hafalla	Project Manager	04/17/26

1. Introduction

1.1 Purpose

1.2 Scope

1.3 Definitions, Acronyms, and Abbreviations

1.4 References

2. Specific Requirements

2.1 External Interface Requirements

2.1.1 User Interfaces

2.1.2 Hardware Interfaces

2.1.3 Software Interfaces

2.1.4 Communications Interfaces

3.1 Front End Details

3.2 Technical Requirements

3.3 Functional validations

3.4 Classes / Objects

3.5 Non-Functional Requirements

3.7 Design Constraints

3.8 Logical Database Requirements

3.9 Other Requirements

4. Analysis Models

5. Change Management Process

1. Introduction

The **Marikina LegisHub** has its goal to manage legislative documents within the City Council of Marikina.

1.1 Purpose

*The purpose of this document is to define the functional and non-functional requirements for the **Marikina LegisHub**, a digital legislative document management system. This SRS serves as the primary blueprint for the development team, quality assurance testers, and stakeholders within the Marikina City Council to ensure the platform meets the city's governance and data integrity standards.*

1.2 Scope

The Marikina LegisHub is a web-based repository designed to automate the filing, tracking, and retrieval of city ordinances and resolutions.

In-Scope:

- **Centralized Repository:** Digital storage of all legislative records with categorized indexing.
- **Search & Retrieval:** Advanced filtering by date, author, and legislative category.
- **User Management:** Role-based access control for Admins (City Secretary), Legislators, and Citizens.
- **Audit Trail:** Logging of all document modifications to ensure data integrity.
- **Responsive Web Interface:** Compatibility with modern web browsers (Chrome, Firefox, Edge, Safari).

Out-of-Scope:

- **Legacy Data Conversion:** Manual encoding of physical paper archives predating the system's launch is not included in the initial development phase.
- **Offline Access:** The system requires an active internet connection or local government network (Intranet) access.
- **Mobile App Development:** While the website will be mobile-responsive, a native Android/iOS application is currently beyond the scope.

1.3 Definitions, Acronyms, and Abbreviations

Abbreviation	Word
RBAC	Role-Based Access Control
OCR	Optical Character Recognition
SMTP	Simple Mail Transfer Protocol
LDMS	Legislative Document Management System

1.4 References

Manoharan, A. P., Melitski, J., & Holzer, M. (2022). *Digital governance: An assessment of performance and readiness of municipal government websites.* *Public Management Review*, 24(3), 389–406. <https://doi.org/10.1007/s11115-021-00584-8>

Xu, Y., & Dai, J. (2024). *Evaluation of local government digital governance ability: A digital governance capacity indicator system.* *Sustainability*, 16(4), 1654. <https://doi.org/10.3390/su16041654>

David, A., Yigitcanlar, T., Li, R. Y. M., Corchado, J. M., Cheong, P. H., Mossberger, K., & Mehmood, R. (2023). *Understanding local government digital technology adoption strategies: A PRISMA review.* *Sustainability*, 15(12), 9645. <https://doi.org/10.3390/su15129645>

Maneja, C. P., Tandang, N. A., & Paunlagui, M. M. (2012). *Determinants in sustaining a local information system in the Philippines: The case of the Barangay Management Information System (BMIS).* *Journal of Public Affairs and Development*, 1(2), 119–139. <https://www.ukdr.uplb.edu.ph/jpad/vol1/iss2/4>

Philippine News Agency. (2023, September 13). *Baguio launches online legislative archive for transparency.* Philippine News Agency. <https://www.pna.gov.ph/articles/1209662>

Department of the Interior and Local Government – CALABARZON. (2024, April 8). *DILG launches Document Management System for LGUs (DMS4LGUs).* Department of the Interior and Local Government Region IV-A CALABARZON.

<https://calabarzon.dilg.gov.ph/news/dilg-launches-document-management-system-for-lgus-dms4lgus>

Davis, F. D. (1989). *Perceived usefulness, perceived ease of use, and user acceptance of information technology.* *MIS Quarterly*, 13(3), 319–340. <https://doi.org/10.2307/249008>

DeLone, W. H., & McLean, E. R. (2003). *The DeLone and McLean model of information systems success: A ten-year update.* *Journal of Management Information Systems*, 19(4), 9–30. <https://doi.org/10.1080/07421222.2003.11045748>

DeLone, W. H., & McLean, E. R. (2016). *Information systems success measurement.* *Foundations and Trends® in Information Systems*, 2(1), 1–116. <https://doi.org/10.1561/29000000005>

Kala, D., Titiyal, R., & Chaubey, D. S. (2024). *Adoption of e-government services: An integrated model approach.* *International Journal of Public Administration*. [Advance online publication]. <https://doi.org/10.1080/01900692.2024.2301234>

Kofahi, I. (2020). *The effect of information systems success model on the user satisfaction of e-government services.* *International Journal of Technology Diffusion*, 11(3), 1–16. <https://doi.org/10.4018/IJTD.2020070101>

Mailizar, M., Burg, D., & Maulina, S. (2021). *Examining the intention to use e-learning in a high-tech environment: An integration of TAM and IS success model.* *Education and Information Technologies*, 26, 3551–3572. <https://doi.org/10.1007/s10639-020-10406-y>

Petter, S., DeLone, W., & McLean, E. R. (2013). *Information systems success: The quest for the independent variables.* *Journal of Management Information Systems*, 29(4), 7–62. <https://doi.org/10.2753/MIS0742-1222290401>

Stefanovic, D., Marjanovic, U., Delic, M., Culibrk, D., & Lalic, B. (2016). Assessing the success of e-government systems: An employee perspective. *Information Systems Frontiers*, 18(2), 281–297. <https://doi.org/10.1007/s10796-015-9577-y>

Venkatesh, V., & Davis, F. D. (2000). A theoretical extension of the technology acceptance model: Four longitudinal field studies. *Management Science*, 46(2), 186–204. <https://doi.org/10.1287/mnsc.46.2.186.11926>

Wang, Y. S., & Liao, Y. W. (2008). Assessing e-government systems success: A validation of the DeLone and McLean model of information systems success. *Government Information Quarterly*, 25(4), 717–733. <https://doi.org/10.1016/j.giq.2007.06.002>

Widyaningrum, P. W., Puspitasari, D., & Wardhana, A. C. (2024). Analysis of the effect of service quality on user satisfaction of the Merdeka Mengajar platform using the DeLone and McLean method. *Jurnal Ilmiah Teknik Elektro Komputer dan Informatika (JITEKI)*, 10(1), 12–22. <https://doi.org/10.23887/jet.v8i1.71080>

Wirtz, B. W., & Piper, M. (2016). Open government: The role of service quality, satisfaction, and acceptance. *Public Administration Quarterly*, 40(4), 837–873. <https://doi.org/10.26466/opusjsr.1296596>

International Organization for Standardization. (2016). Information and documentation – Records management (ISO 15489). ISO. <https://www.iso.org/standard/62542.html>

Office of the President of the Philippines. (1987). Executive Order No. 292: Administrative Code of 1987. Official Gazette of the Republic of the Philippines. <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/5/53270>

Commission on Audit. (2023). Internal Audit Manual (IAM) for Local Government Units (LGUs) (2023 Edition). Government of the Philippines. <https://www.dbm.gov.ph/wp-content/uploads/Issuances/2023/Local-Budget-Circular/IAM-for-LGUs-2023-Edition.pdf>

Reyes, D. D. (2009). The Local Government Code and the challenges of local autonomy in the Philippines. *Philippine Journal of Public Administration*, 53(3-4), 212–235.

<https://cids.up.edu.ph/wp-content/uploads/2023/03/Assessing-Local-Governance-and-Autonomy-in-the-Philippines.pdf>

Development Academy of the Philippines. (2018). Towards a Requisite Regulatory Management System: Philippines. Development Academy of the Philippines. <https://pdc.dap.edu.ph/index.php/institutionalization-of-regulatory-management-system/>

United Nations. (n.d.). Goal 9: Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation. United Nations Sustainable Development. <https://sdgs.un.org/goals/goal9>

United Nations. (n.d.). Goal 11: Make cities and human settlements inclusive, safe, resilient and sustainable. United Nations Sustainable Development. <https://sdgs.un.org/goals/goal11>

United Nations. (n.d.). Goal 12: Ensure sustainable consumption and production patterns. United Nations Sustainable Development. <https://sdgs.un.org/goals/goal12>

United Nations. (n.d.). Goal 16: Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels. United Nations Sustainable Development. <https://sdgs.un.org/goals/goal16>

United Nations. (n.d.). Goal 17: Strengthen the means of implementation and revitalize the Global Partnership for Sustainable Development. United Nations Sustainable Development. <https://sdgs.un.org/goals/goal17>

United Nations. (2015). Transforming our world: The 2030 Agenda for Sustainable Development. Department of Economic and Social Affairs. <https://sdgs.un.org/goals>

2. Specific Requirements

The Proposed Project will have 2 roles

1. System Administrator (City Secretary / IT Staff)
2. Council Member (Legislators / Department Heads)

Following features/modules will be available to these 2 different roles

System Administrator	Council Member (Legislator)
Full System Configuration	Search & View Ordinances
Upload & Index New Documents	Track Resolution Status
Edit/Correct Metadata	Download Official Copies
Manage User Accounts & Permission	Personal Bookmarking/Saved Docs
Permanent Record Deletion	View Legislative History
View System-Wide Audit Logs	Change Personal Password
Login & Logout	Login & Logout

Description of the modules

Module Name	Applicable Roles	Description
Login & Logout	System Admin, Council Member	System Administrator: Can log in with full administrative access to oversee the entire system and manage security settings. Terminating the session ensures high-level access tokens are cleared. Council Member: Can log in with restricted access focused on document processing and record management. Logging out safely ends the session to prevent unauthorized terminal access.
Full System Configuration	Admin Only	System Administrator: Grants authority to modify global system parameters, including metadata categories, document types, and department settings to ensure the system aligns with city hall standards.
Upload & Index New Documents	Admin Only	System Admin: Provides the interface for digitizing new legislative records by entering primary metadata and uploading the official PDF file to the centralized digital repository.

Edit/Correct Metadata	Admin Only	System Admin: Allows for the correction of typographical errors or updates to existing record details (such as dates or authors) to maintain the integrity and accuracy of the digital archive.
Manage User Accounts & Permissions	Admin Only	System Admin: A security-focused module used to create new accounts, assign specific roles, and manage permission levels for all City Council personnel.
Permanent Record Deletion	System Admin, Council Member	System Admin: Reserved for authorized IT staff to permanently remove redundant or erroneous records from the system database, ensuring the repository remains clean and relevant.
View System-Wide Audit Logs	System Admin	System Admin: Monitors a comprehensive, non-editable trail of all system activities, including user actions, timestamps, and module changes for high-level accountability.
Search & View Ordinances	Council Member	Council Member: Provides an optimized search interface to quickly locate and preview local ordinances by title, year, or keywords to facilitate legislative research.
Track Resolution Status	Council Member	Council Member: Allows legislators to monitor the current status of specific resolutions (e.g., Pending, Approved, or Archived) in real-time.

Download Official Copies	Council Member	Council Member: Enables the retrieval of official PDF copies of legislative documents for use in council sessions, committee hearings, or public reference.
Personal Bookmarking/Saved Docs	Council Member	Council Member: Provides a personalized workspace where legislators can "save" or bookmark frequently accessed documents for faster retrieval in future sessions.
View Legislative History	Council Member	Council Member: Allows users to track the timeline and version history of a specific document, showing how it has evolved from its initial filing to its final enactment.
Change Personal Password	Council Member	Council Member: A self-service security module that allows legislators to update their own login credentials periodically to ensure their account remains secure.

2.1 External Interface Requirements

2.1.1 User Interfaces

Interface/UI Component	Module	Description
Login Screen	Authentication	The main entry portal featuring text fields for Email and Password, a "Sign In" button, and links for password recovery.

Admin Dashboard	Dashboard	A high-level overview page displaying statistical cards for Total Ordinances, Total Resolutions, and Active User Accounts.
Summary Cards	Dashboard	Visual indicators at the top of the dashboard providing real-time counts of the system's total legislative records.
Pending Documents Table	Dashboard	A list view on the dashboard showing recently uploaded documents that require immediate review or final indexing.
Main Navigation Sidebar	Global	A persistent left-side menu containing links to Dashboard, Manage Documents, User Management, Activity Log, and Settings.
Manage Documents Page	Document Management	The central repository table listing all legislative files with columns for ID, Title, Type, and Date Enacted.
Add Document Modal	Document Management	A pop-up form with input fields for Document Type, Number, Date, Title, and Author/Sponsor for creating new records.
File Upload Zone	Document Management	A drag-and-drop or file-picker area within the Add Document modal for attaching official PDF copies.
Document Details View	Document Management	A dedicated page/modal showing the full metadata of a specific ordinance or resolution alongside a PDF previewer.
Legislative Remarks Tab	Document Management	A sub-section within the document view where Council Members can read or add notes regarding a document's history.

Edit Metadata Modal	Document Management	A pre-populated form that allows Administrators to correct or update details of an existing legislative record.
Advanced Search Bar	Search	A dynamic search field at the top of document lists that filters results in real-time as the user types.
Filter Dropdown Menu	Search	A multi-select menu allowing users to narrow down documents by Year, Category, or Status.
User Management Table	User Management	A comprehensive list of all system users displaying their Full Name, Email, Assigned Role, and Account Status.
Add User Modal	User Management	A form for Administrators to register new personnel, including fields for Name, Role Selection, and temporary credentials.
Role Selection Toggle	User Management	A dropdown or radio button group within user forms to switch between "System Administrator" and "Council Member."
Edit User Profile Modal	User Management	An interface to modify existing staff accounts, such as updating their role or toggling their status to "Inactive."
Audit Log Table	Activity Log	A detailed chronological list showing User ID, Action Taken (e.g., "Updated Metadata"), Module, and the exact Timestamp.
Archive Repository Page	Archive	A specialized view for documents that have been removed from the main list but are preserved for legal purposes.

Retention Status Indicator	Archive	A visual tag in the archive table showing how long a document has been stored or its scheduled disposal date.
Restore Confirmation Modal	Archive	A warning prompt that appears when an Admin attempts to move a document from the Archive back to the active list.
Delete Permanently Prompt	Archive	A high-security confirmation modal for the permanent purging of erroneous records from the database.
Profile Settings Interface	Settings	A user-specific page for managing account details like display name and profile picture.
Password Update Form	Settings	A secured section within Settings with fields for "Current Password," "New Password," and "Confirm New Password."
System Configuration Panel	Settings	An Admin-only view for adjusting global settings, such as backup frequencies and system-wide notifications.

2.1.2 Hardware Interfaces

Category	Hardware Component	Minimum Specifications	Description / Justification
----------	--------------------	------------------------	-----------------------------

Computer Hardware (Server)	Processor (CPU)	Intel Core i5-10400 (6 Cores, 12 Threads) or higher	Acts as the central brain of the system; necessary for handling concurrent requests from different City Council departments and managing database queries.
	Memory (RAM)	16GB DDR4 3200MHz	Ensures high-speed performance when indexing large PDF files and running the web server without latency.
	Storage (SSD/HDD)	1TB NVMe SSD	Provides fast read/write speeds for the digital repository. SSD is required to ensure that document retrieval takes less than 3 seconds.
Computer Hardware (Client)	Workstation (PC/Laptop)	Intel Core i3 or equivalent, 8GB RAM	Standard office workstations used by the Admin and Councilors to access the web-based dashboard via a browser.
	Monitor/Display	1920 x 1080 (Full HD) Resolution	Necessary for the clear viewing of legislative documents and side-by-side comparison of document versions.

Network & Connectivity	Network Interface Card (NIC)	10/100/1000 Mbps Gigabit Ethernet	Facilitates rapid data transfer between the client workstations and the central server, especially for large document uploads.
	Router / Switch	Dual-Band Wi-Fi 6 Router or Managed Switch	Provides a stable and secure local network environment within the Marikina City Hall for system accessibility.
	Internet Connection	50 Mbps Fiber Connection	Required for cloud-based deployment or remote access by Council Members during off-site legislative reviews.
Peripherals	Document Scanner	High-Speed ADF (Auto Document Feeder) Scanner	Essential hardware for the System Administrator to convert physical paper-based ordinances into digital PDF formats for system entry.
	Uninterruptible Power Supply (UPS)	650VA / 360W UPS	Protects the server hardware from power surges and prevents data corruption during Marikina's frequent environmental challenges (e.g., power outages during floods).

	External Storage	2TB External Hard Drive	Used for secondary physical backups to comply with the project's goal of data integrity and disaster recovery.
--	-------------------------	-------------------------	--

2.1.3 Software Interfaces

Category	Software Component	Specifications	Description / Justification
Operating System	Windows 10 / 11	64-bit Architecture	The primary operating system for both the server deployment and client-side workstations within the Marikina City Hall environment.
Database Management System (DBMS)	MySQL	Version 8.0 or higher	A robust relational database used to store legislative metadata, user credentials, and non-editable audit trails.
Backend Framework	Python (Django / FastAPI)	Python 3.10+	Used to build the core logic of the system, handling Role-Based Access Control (RBAC), file indexing, and secure communication between the UI and the database.

Web Browser	Google Chrome, Mozilla Firefox, Microsoft Edge	Latest Stable Versions	The system is optimized for Chromium-based and Gecko-based browsers to ensure all Figma-designed UI elements and modals render correctly.
PDF Libraries	PyMuPDF / ReportLab / PDF.js	Integration Libraries	Essential software libraries used to process, view, and generate official PDF copies of ordinances and resolutions directly within the web interface.

2.1.4 Communications Interfaces

Category	Protocol / Interface	Specifications	Description / Justification
Network Addressing	Domain Name System (DNS)	legishub.marikina.gov.ph	The system is hosted on a dedicated government sub-domain rather than a local IP address. This allows users across different departments to access the portal using a standard, memorable web URL via the city's network.

Data Transfer	HTTP / HTTPS	TLS 1.2 / 1.3 Encryption	All traffic between the end-user's browser and the web server is encrypted using SSL/TLS certificates. This protects sensitive legislative data and user credentials from interception during transmission.
Network Scope	WAN & Intranet	City-wide Network Infrastructure	The system utilizes the Marikina City Government's Wide Area Network (WAN), ensuring that the server is reachable from various departments and legislative offices outside the primary server room.
Secure File Handling	SFTP (Secure File Transfer Protocol)	SSH-based Tunneling	Provides a secure encrypted tunnel for the application to move uploaded PDF ordinances and resolutions from the client interface to the protected backend storage directory.
Alerting System	SMTP (Simple Mail Transfer Protocol)	Port 465 or 587 (SSL/TLS)	Utilized by the Time-Based Review & Escalation Module to send automated email notifications and alerts to City Council staff regarding pending document reviews or status changes.

3.1 Front End Details

This section describes the specific data fields required for each module of the **Marikina LegisHub**. These fields ensure standardized data entry for all city ordinances and resolutions.

Global Navigation (Sidebar)

- Dashboard (Link)
- Manage Documents (Link)
- User Management (Link)
- Activity Log (Link)
- Archive (Link)
- Settings (Link)
- Logout (Link)

Login Page

- Email Address (Input Field)
- Password (Masked Input Field)
- Keep me logged in (Checkbox)
- Forgot Password? (Link)
- Sign In (Button)

Dashboard Interface

- Total Ordinances (Card Value)
- Total Resolutions (Card Value)
- Active User Accounts (Card Value)
- Pending Documents (Table View)
- Document Title (Table Column)
- Type (Table Column)
- Date Uploaded (Table Column)
- Quick View (Action Button)

Manage Documents (Main Table)

- Add New Document (Primary Action Button)
- Search Documents (Input Field)
- Export to Excel/PDF (Button)
- Document ID (Table Column)
- Title (Table Column)
- Document Type (Table Column)

- Author/Sponsor (Table Column)
- Date Enacted (Table Column)
- Status (Table Column: e.g., Pending, Approved)
- Actions (View, Edit, Archive Icons)

Add New Document (Modal)

- Document Type (Dropdown: Ordinance, Resolution, Memorandum)
- Document Number (Input Field)
- Date Enacted (Date Picker)
- Document Title (Text Area)
- Author/Sponsor (Input Field)
- Tags/Keywords (Input Field)
- Select PDF File (File Upload/Drag & Drop)
- File Name Preview (Label)
- Submit (Button)
- Cancel/Reset (Button)

Edit/Correct Metadata (Modal)

- Document ID (Read-only)
- Document Type (Dropdown)
- Document Number (Input Field)
- Date Enacted (Date Picker)
- Document Title (Text Area)
- Author/Sponsor (Input Field)
- Status (Dropdown: Pending, Approved, Revoked)
- Replace PDF File (Optional Upload)
- Update Changes (Button)
- Close (Button)

Advanced Search & Filtering

- Search by Keyword (Input)
- Category (Dropdown: All, Ordinance, Resolution)
- Year (Dropdown/Date Picker)
- Proponent/Author (Dropdown/Input)
- Filter (Button)
- Clear All Filters (Link)

User Management (Main Table)

- Add New User (Primary Action Button)
- User ID (Table Column)

- Full Name (Table Column)
- Email Address (Table Column)
- Role (Table Column: Admin, Council Member)
- Account Status (Table Column: Active, Inactive)
- Date Created (Table Column)
- Edit User (Action Icon)

Add New User (Modal)

- First Name (Input Field)
- Last Name (Input Field)
- Email Address (Input Field)
- Role (Dropdown: System Admin, Council Member)
- Set Temporary Password (Input Field)
- Profile Picture Upload (Optional)
- Submit (Button)
- Reset (Button)

Edit/Update User (Modal)

- Full Name (Read-only/Edit)
- Email Address (Input Field)
- Change Role (Dropdown)
- Account Status (Toggle/Dropdown: Active/Inactive)
- Reset Password (Button)
- Save Changes (Button)
- Cancel (Button)

Activity / Audit Log Table

- Log ID (Table Column)
- User Name (Table Column)
- Action Performed (Table Column: e.g., "Deleted Document", "Logged In")
- Module Affected (Table Column)
- Timestamp (Table Column: Date & Time)
- IP Address (Table Column - for Security)
- View Details (Action Link)

Archive Repository

- Search Archive (Input Field)
- Date of Archival (Table Column)
- Retention Status (Table Column)
- Restore to Active (Button Icon)

- Delete Permanently (Red Action Button)
- Bulk Restore (Button)

Document View & History (Council Member)

- Document Previewer (PDF Display Pane)
- Version History (Sidebar List)
- Proponent Name (Label)
- Date Passed (Label)
- Current Status (Badge)
- Legislative Remarks (Text Area)
- Post Remark (Button)
- Download PDF (Button)
- Bookmark Document (Toggle/Icon)

Profile Settings

- First Name (Input)
- Last Name (Input)
- Email Address (Input)
- Display Picture (Image Upload)
- Save Profile (Button)

Security Settings (Change Password)

- Current Password (Masked Input)
- New Password (Masked Input)
- Confirm New Password (Masked Input)
- Password Strength Meter (Visual Indicator)
- Change Password (Button)
- Reset Form (Button)

3.2 Technical Requirements

Document Registration (Ordinances/Resolutions)

T1 Ordinance/Resolution No – Field must not be blank

T2 Ordinance/Resolution No – Special characters (except hyphens) are restricted

T3 Ordinance/Resolution No – First character cannot be a space

T4 Document Title – Document Title must not be blank

T5 Document Title – Special characters are restricted to prevent SQL injection

T6 Document Title – First character cannot have a space

T7 Author/Sponsor – Author/Sponsor must not be blank

- T8** Author/Sponsor – Numbers are not allowed
- T9** Author/Sponsor – Special characters are not allowed
- T10** Author/Sponsor – First character cannot have a space
- T11** Date Enacted – Date Enacted field must not be blank
- T12** Date Enacted – Future dates are not allowed
- T13** PDF Attachment – Only .pdf extensions are allowed
- T14** PDF Attachment – File size must not exceed 25MB
- T15** PDF Attachment – Corrupted files are not allowed

Staff/User Profiles

- T16** Staff Name – Staff name must not be blank
- T17** Staff Name – Numbers are not allowed
- T18** Staff Name – Special characters are not allowed
- T19** Staff Name – First character cannot have a space
- T20** Office/Department – Selection must not be blank
- T21** Telephone/Extension – Mobile/Ext no must not be blank
- T22** Telephone/Extension – Must be numeric only
- T23** Telephone/Extension – Special characters are not allowed
- T24** Telephone/Extension – Must be at least 7 to 11 digits
- T25** Telephone/Extension – First character cannot have a space
- T26** Email – Email ID must not be blank
- T27** Email – Email format is not valid
- T28** Email – First character cannot have a space
- T29** User Role – Role selection is required

Legislative Search & Retrieval

- T30** Search Query – Search query must not be blank
- T31** Search Query – Special characters are filtered/escaped
- T32** Year Filter – Must be a 4-digit numeric value
- T33** Year Filter – Characters are not allowed
- T34** Year Filter – Special characters are not allowed
- T35** Document Type Filter – Selection must not be blank

System Login

- T36** User-ID – User-ID must not be blank
- T37** User-ID – Special characters are not allowed
- T38** User-ID – Spaces are not allowed
- T39** Password – Password must not be blank
- T40** Password – Characters must be masked

Change Password

- T41** Old Password – Old Password must not be blank
- T42** New Password – New Password must not be blank
- T43** New Password – Enter at-least one numeric value
- T44** New Password – Enter at-least one special character
- T45** New Password – Password must be at least 8 characters
- T46** Confirm Password – Confirm Password must not be blank
- T47** Confirm Password – Passwords do not match

Time Escalation & Audit Logic

- T48** Escalation Threshold – Threshold Days must not be blank
- T49** Escalation Threshold – Characters are not allowed
- T50** Escalation Threshold – Special characters are not allowed
- T51** Escalation Threshold – Negative values are not allowed
- T52** Reason for Edit – Description text box must not be blank
- T53** Reason for Edit – First character cannot have a space
- T54** Audit Log Integrity – Client-side clock tampering is restricted
- T55** Audit Log – Log entries are read-only

Archive & Disposal

- T56** Archive Action – Confirmation modal is required
- T57** Permanent Deletion – Master Admin Password is required
- T58** Restoration – Duplicate Document Number check is required
- T59** Archival Metadata – System-generated timestamp is required
- T60** Bulk Actions – Selected items must not be in "Locked" status

3.3 Functional validations

Document Upload / Edit Form

System Admin

- F1** Ordinance/Resolution Number: Field must not be blank.
- F2** Ordinance/Resolution Number: Special characters (except hyphens -) are not allowed.
- F3** If the Ordinance/Resolution number already exists in the database, the system displays an error.
- F4** Title: Field must not be blank.
- F5** Title: First character cannot be a space.
- F6** Proponent: Must be selected from the predefined list of Council Members.
- F7** Date of Passage: Must be a valid date and cannot be a future date.
- F8** File Upload: File must be in .pdf or .docx format.
- F9** File Upload: File size must not exceed 25MB.
- F10** If a document is modified, the description field in the audit log must not be blank.

User Management (Admin Only) System

Admin

- F11** Full Name: Numbers and special characters are not allowed.
- F12** Full Name: Field must not be blank.
- F13** Email: Must follow a valid format (e.g., name@marikina.gov.ph).
- F14** Email: First character cannot be a space.
- F15** If the same Email Id exists in the system, the system shows an error.
- F16** Username: Special characters and spaces are not allowed.
- F17** User Role: A role (Admin or Council Member) must be selected.
- F18** If the Admin attempts to delete a User ID that does not exist, the system displays an error.

Search & Filter System

Admin

- F19** Keyword Search: If used, must contain at least 3 characters to prevent broad queries.
- F20** Year Filter: Must be a 4-digit numeric value (e.g., 2026).
- F21** Date Range: "To Date" cannot be earlier than "From Date."
- F22** System Admin can search and view all documents including archived and restricted files.

Council Member

- F23** Keyword Search: If used, must contain at least 3 characters.
- F24** Year Filter: Must be a 4-digit numeric value.
- F25** Date Range: "To Date" cannot be earlier than "From Date."
- F26** Council Member can only search for and view official ordinances and resolutions approved for their access level.
- F27** If a Council Member attempts to access a document ID that is restricted, the system displays an error.

Time Escalation Settings

System Admin

- F28** Threshold Days: Must be a numeric value; characters are not allowed.
- F29** Threshold Days: Value must be greater than 0.
- F30** Notification Email: Must be a valid email address.
- F31** If the notification email field is left blank, the system displays an error.

Login & Security

System Admin

- F32** User-ID: Must not be blank.
- F33** Password: Must not be blank.
- F34** If the Admin User-ID or Password does not exist in the database, the system displays an error.

Council Member

- F35** User-ID: Must not be blank.
- F36** Password: Must not be blank.
- F37** If the Council Member credentials are invalid, the system displays an error.

Change Password

System Admin

F38 If Old Password does not match the current database record, the system displays an error.

F39 New Password: Must contain at least 8 characters.

F40 New Password: Must contain at least one numeric value and one special character.

F41 Confirm Password: Must match the "New Password" field exactly; otherwise, system displays an error.

Council Member

F42 If Old Password is invalid, the system displays an error.

F43 New Password must follow the complexity rules (8 characters, numeric, and special character).

F44 Confirm Password must match exactly or the system displays an error.

Audit & History System

Admin

F45 Reason for Edit: Special characters are restricted to prevent SQL injection.

F46 If an Admin attempts to modify a record without providing a reason, the system displays an error.

Archive & Disposal

System Admin

F47 If a document is moved to archive, it must exist in the active records database.

F48 Permanent Deletion: If the secondary Admin password is incorrect, the system displays an error.

F49 If a document is being restored but its original Document Number is now taken by another record, the system displays an error.

3.4 Classes / Objects

This section describes the core domain objects of **Marikina LegisHub** as defined in the database schema and Functional Specification Document. Each class is presented with its purpose, key attributes (mapped directly to database fields), key relationships, and primary behaviors.

3.5.1.1 Attributes

USER ACCOUNT

Represents a unified system account for all internal personnel. Role-based attributes differentiate access levels between the System Administrator and Council Members.

Key Attributes:

- **id** (PK, auto-generated integer)
- **username** (unique, required)
- **email** (unique, required – @marikina.gov.ph format)

- **password** (hashed, required)
- **first_name, last_name** (required)
- **is_staff** (boolean – TRUE for System Admin, FALSE for Council Members)
- **is_active** (boolean – FALSE blocks system access)
- **last_login** (auto-recorded timestamp)
- **date_joined** (auto-recorded at registration)

Key Relationships:

- One-to-many with **LegislativeDocument** (as the uploader/author)
 - One-to-many with **AuditLog** (as the actor performing system activities)
 - One-to-many with **Notification** (as the recipient of alerts)
- ### Document and Legislative Domain

LEGISLATIVE DOCUMENT

Represents the core legislative records (Ordinances and Resolutions). It captures legal metadata, physical and digital storage tracking, and visibility status.

Key Attributes:

- **id** (PK, auto-generated integer)
- **document_number** (Unique – e.g., "2026-ORD-01")
- **title** (required, SQL-injection protected)
- **doc_type** (Ordinance, Resolution)
- **sponsor** (required – maps to Council Member list)
- **year** (4-digit integer for series filtering)
- **date_enacted** (Date – cannot be a future date)
- **date_filed** (Date – entry timestamp)
- **status** (Pending, Approved, Revised, Archived)
- **visibility** (Public, Restricted, Internal Only)
- **keywords** (Tags for metadata-based searching)
- **file_attachment** (file – PDF only, max 25MB)
- **physical_storage** (string – floor/cabinet location for physical backup)
- **uploaded_by_id** (FK – references [User.id](#))

Key Relationships:

- Belongs to **UserAccount** (via uploaded_by_id)
- One-to-many with **AuditLog** (tracks version history and edits)
- One-to-one with **ArchiveRecord** (when moved to archival status)

AUDIT LOG

Ensures accountability by capturing every significant action performed within the system, including data modifications, deletions, and login events.

Key Attributes:

- **id** (PK, auto-generated integer)
- **timestamp** (Server-side generated clock)
- **user_id** (FK – references UserAccount.id)
- **document_id** (FK – references LegislativeDocument.id)
- **action** (LOGIN, CREATE, UPDATE, DELETE, ARCHIVE, PURGE)
- **description** (Text – includes the "Reason for Edit")
- **ip_address** (captured for security tracking)

Key Relationships:

- Belongs to **UserAccount** (identifies the person responsible)
- Associated with **LegislativeDocument** (tracks the lifecycle of a specific record)

SYSTEM CONFIGURATION

Manages global system settings, cloud synchronization, and disaster recovery configurations.

Key Attributes:

- **city_hall_name** (String)
- **office_department** (String)
- **backup_frequency** (Daily / Weekly)
- **cloud_sync_status** (Boolean)
- **last_handshake_time** (Timestamp)
- **integrity_health_score** (Percentage)

Key Relationships:

- One-to-many with **AuditLog** (logs configuration changes)

SECURITY POLICY

Manages access control rules and network security protocols.

Key Attributes:

- **session_timeout_limit** (Integer - Minutes)
- **ip_restriction_enabled** (Boolean)
- **mfa_required** (Boolean)
- **password_expiry_days** (Integer)

ARCHIVE RECORD

Represents documents that have reached their active lifecycle end but are retained for legal compliance.

Key Attributes:

- **id** (PK, auto-generated integer)
- **archive_id** (Unique – ARC-XX-XXX)

- **original_document_id** (FK – references LegislativeDocument.id)
- **date_archived** (Date)
- **retention_status** (Permanent / Years Left)
- **archived_by_id** (FK – references [UserAccount.id](#))

Key Relationships:

- Belongs to **LegislativeDocument** (original reference)
- Belongs to **UserAccount** (the official who authorized the archival)

3.5.1.2 Functions

This section describes the primary behaviors and technical operations (methods) associated with each class. These functions define how the system processes data, enforces security, and manages the legislative lifecycle of documents in **Marikina LegisHub**.

Identity and Access Domain

USER ACCOUNT

Manages the lifecycle of user sessions and account security.

Primary Functions:

- **Authenticate():** Validates the **username** and **password** against the hashed values in the database. Returns a session token upon success.
- **EnforceRoleAccess():** Checks the **is_staff** attribute to determine if the user can access administrative modules like User Management or System Settings.
- **UpdateProfile():** Allows users to modify their **first_name**, **last_name**, and **email**. Triggers a validation check for the @marikina.gov.ph format.
- **ResetPassword():** Validates the **old_password** and updates the **password** field with a new hashed value, ensuring it meets length and complexity requirements.
- **ToggleStatus():** (Admin only) Switches the **is_active** boolean to enable or disable an account's ability to log in.

Document and Legislative Domain

LEGISLATIVE DOCUMENT

Handles the creation, retrieval, and modification of ordinances and resolutions.

Primary Functions:

- **RegisterMetadata():** Saves a new record to the database. It validates that the

`document_number` is unique and the `date_enacted` is not in the future.

- **UploadFile():** Processes the `file_attachment`. It verifies the file is a PDF, checks that it is under 25MB, and assigns a secure storage path.
- **SearchDocuments():** Executes a query based on `keywords`, `year`, `doc_type`, or `sponsor`. It utilizes indexing on the `title` and `keywords` for performance.
- **ModifyRecord():** Updates existing attributes. It requires a "Reason for Edit" input which is then passed to the `AuditLog` class.
- **InitiateArchival():** Changes the `status` to "Archived" and triggers the creation of a corresponding `ArchiveRecord`.

AUDIT LOG

Provides a transparent, unalterable trail of system activities.

Primary Functions:

- **GenerateEntry():** Automatically triggered by other functions (e.g., `UploadFile`, `Login`). It captures the `timestamp`, `user_id`, and `ip_address` at the moment of the action.
- **FetchHistory():** Retrieves logs filtered by `user_id` or `document_id` to show the chronological history of a specific entity.
- **PurgeOldLogs():** (System only/Admin triggered) Deletes entries where the `timestamp` is older than 5 years to optimize database storage.

SYSTEM CONFIGURATION

Governs the operational environment and data safety.

Primary Functions:

- **SyncToCloud():** Executes a "handshake" with the Marikina Secure Cloud to upload database backups. Updates `last_handshake_time`.
- **CalculateIntegrityScore():** Compares local file hashes with cloud backups to determine the `integrity_health_score`.
- **UpdateSettings():** Modifies `backup_frequency` and city hall metadata.

SECURITY POLICY

Automated enforcement of network and session security.

Primary Functions:

- **MonitorInactivity():** Tracks the time since the `last_login` or last activity. Triggers a logout if the `session_timeout_limit` is exceeded.
- **ValidateIP():** If `ip_restriction_enabled` is TRUE, it compares the current `ip_address`

against the allowed Marikina City Hall local network range.

ARCHIVE RECORD

Manages the long-term retention and potential restoration of files.

Primary Functions:

- **CalculateRetention():** Computes the **retention_status** based on the **date_archived** and predefined legal schedules.
- **RestoreDocument():** Reverts an **ArchiveRecord** back into a **LegislativeDocument** and moves it from the archive module back to the active list.
- **PermanentPurge():** Deletes the record and the associated physical file path after secondary admin confirmation.

3.5 Non-Functional Requirements

3.1 Security

- **Encrypted Domain Access:** The system must be hosted on a domain with a valid **SSL/TLS certificate** (HTTPS) to protect legislative data from "Man-in-the-Middle" attacks.
- **Role-Based Access Control (RBAC):** Access to the Admin Dashboard (for document uploads and user management) must be restricted strictly to authorized City Secretariat personnel.
- **Credential Protection:** All user passwords must be encrypted using industry-standard hashing algorithms (e.g., Argon2 or BCrypt) before being stored in the database.

3.2 Performance

- **Search Response Time:** Under normal network conditions, the system should return search results for ordinances or resolutions in **less than 2 seconds**.
- **Concurrent Access:** The system must handle at least **30 simultaneous users** (Council members and staff) without a noticeable delay in page loading.
- **Document Loading:** The integrated PDF viewer should render documents quickly, even for large files up to **25MB**.

3.3 Reliability

- **System Uptime:** The LegisHub must maintain **99.9% availability** during official Marikina City Hall working hours to ensure it is ready for legislative sessions.
- **Fault Tolerance:** In the event of a server error, the system must display a user-friendly

"Maintenance" page rather than exposing raw code or database errors to the user.

- **Data Backups:** Automated daily backups of the SQL database and the physical PDF storage must be implemented to prevent data loss.

3.4 Usability

- **Responsive Web Design:** The interface must automatically adjust for **Desktop, Tablet, and Mobile** screens, allowing Council Members to view documents on their portable devices.
- **Intuitive Navigation:** The search bar and category filters must be prominently displayed on the landing page for easy access by non-technical staff.
- **Feedback Mechanism:** The system must provide clear notifications (e.g., "Upload Successful" or "Invalid Document Format") to guide the user.

3.5 Data Integrity

- **Uniqueness Constraint:** The system must prevent duplicate entries of the same **Ordinance or Resolution Number** to maintain a "Single Source of Truth."
- **Audit Trail:** Every modification, deletion, or status change of a legislative document must be logged with a **User ID and Timestamp**.
- **Time Escalation Accuracy:** The escalation module must accurately calculate document aging based on the server's system clock to ensure notifications are triggered exactly on schedule.

3.7 Design Constraints

3.7.1 User Interface & Accessibility

- **Intuitive Design:** Recognizing that some users may have limited technical expertise, the system must follow a "Three-Click Rule"—any legislative document should be reachable within three clicks from the homepage.
- **Language:** The primary interface shall be in English, but the search engine must support Filipino keywords commonly found in local ordinances.
- **Visual Hierarchy:** Important actions (like "Upload" for Admins or "Search" for Councilors) must be visually distinct using Marikina City's official color palette to improve recognition.

3.7.2 Software & Browser Standards

- **Cross-Browser Compatibility:** The system must be fully functional on the latest versions of **Google Chrome, Mozilla Firefox, and Microsoft Edge**.

- **Responsive Framework:** The front-end must be built using a responsive framework (e.g., Bootstrap or Tailwind) to ensure the layout adapts to tablets and mobile phones used during sessions.
- **PDF Integration:** The system must utilize a standard web-based PDF viewer (like PDF.js) to ensure documents open within the browser without requiring external software like Adobe Acrobat.

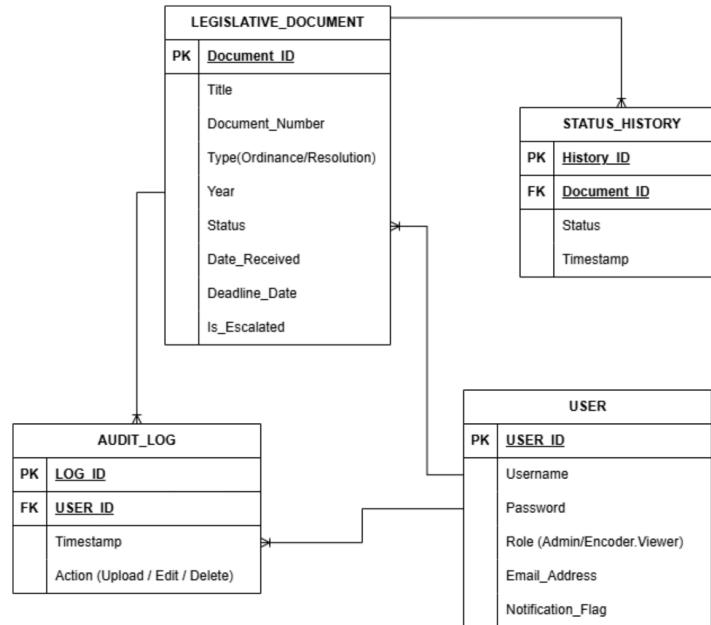
3.7.3 Network & Deployment

- **Domain Hosting:** Per the beneficiary's request, the system must be deployed on a registered **Domain Name** rather than a local IP or `localhost`.
- **Database Constraints:** The system must use a Relational Database Management System (RDBMS) like **MySQL or PostgreSQL** to ensure complex relationships between ordinances, authors, and dates are maintained.
- **Security Standards:** All forms must include CSRF (Cross-Site Request Forgery) protection and input sanitization to prevent unauthorized data injection through the domain.

3.7.4 Regulatory & Legal Constraints

- **Data Privacy:** The system must comply with the **Data Privacy Act of 2012 (RA 10173)** regarding the storage of staff and official information.
- **Document Authenticity:** Digital copies must include a system-generated timestamp or watermark to distinguish them from original physical manuscripts.

3.8 Logical Database Requirements



3.9 Other Requirements

This section lists the external Application Programming Interfaces (APIs), software libraries, and frameworks utilized to build the functional and aesthetic components of the Marikina LegisHub.

Library / API	Purpose	Implementation Snippet
Django Framework	The core high-level Python Web framework used for rapid development and clean, pragmatic design.	<pre>import django, from django.db import models</pre>
Font Awesome API	Used for providing the scalable vector icons for the sidebar, document actions, and dashboard status indicators.	<pre><link rel="stylesheet" href="...font-awesome/6.4.0/css/all. min.css"></pre>

SQLite3 Library	A C-language library that implements a small, fast, self-contained, high-reliability, full-featured, SQL database engine.	DATABASES = { 'default': { 'ENGINE': 'django.db.backends.sqlite3' } }
Google Fonts API	Integrated to provide the 'Segoe UI' and professional sans-serif typography for better readability.	font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
WhiteNoise API	Used for serving static files (CSS, Images) directly from the Django application during deployment.	MIDDLEWARE = ['whitenoise.middleware.WhiteNoiseMiddleware', ...]

4. Analysis Models

Context and data flow diagram

5. Change Management Process

Enumerate some changes you've done at the entire development process

Changes to the SRS either from the development, testing team or the client side will be communicated to the beneficiary.

Any change made to the SRS will require a sign off from the Development lead , QA lead and the client.

Once approved changed will be made to the SRS and the new SRS will be circulated to all stakeholders.

A. Appendices